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ELECTRICITY STORAGE DEVICE

Abstract

An electricity storage device (battery) includes a first cell string (cell string) and a second cell string (cell string). The first cell string and the second cell string each include a plurality of electricity storage cells that is arrayed in a coupling direction and connection parts that electrically connect the electricity storage cells to one another. The electricity storage device further includes wrapping members (members) that are inclined relative to the coupling direction at least at one part and that are wrapped around the electricity storage cells included in the first cell string and the electricity storage cells included in the second cell string.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-026637 filed on Feb. 26, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] This disclosure relates to an electricity storage device including a plurality of cell strings.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication (Translation of PCT Application) No. 2023-502457 (JP 2023-502457 A) discloses a rectangular parallelepiped battery (electricity storage device) of which a length L is 400 mm to 2500 mm and a ratio of the length L to a width H (L/H) is 4 to 21.

SUMMARY

[0004] In the electricity storage device described in JP 2023-502457 A, a plurality of electrode body sets (electricity storage cells) connected in series and arrayed in one row is disposed inside a housing. Hereinafter, a string of a plurality of electricity storage cells connected in one row will be referred to as a “cell string.”

[0005] To improve the volume energy density of an electricity storage device, housing a plurality of cell strings, rather than only one cell string, inside a case is more advantageous. However, if the cell strings move excessively inside the case, a connection part between these cell strings or a connection part between the electricity storage cells is likely to become damaged.

[0006] This disclosure has been made to solve the above-described problem, and an object thereof is to improve the impact resistance of an electricity storage device including a plurality of cell strings.

[0007] According to one form of this disclosure, an electricity storage device shown below is provided.

[0008] (First Item) The electricity storage device includes a first cell string and a second cell string. The first cell string and the second cell string each include a plurality of electricity storage cells that is arrayed in a coupling direction and connection parts that electrically connect the electricity storage cells to one another. The electricity storage device further includes wrapping members that are inclined relative to the coupling direction at least at one part and that are wrapped around the electricity storage cells included in the first cell string and the electricity storage cells included in the second cell string.

[0009] When the electricity storage cells of the first cell string and the electricity storage cells of the second cell string are provided with the wrapping members as described above, the impact resistance of the first cell string and the second cell string is likely to improve. When the wrapping members are wrapped around the electricity storage cells of the respective cell strings so as to be inclined relative to the coupling direction, the first cell string and the second cell string are likely to have high strength against both impact in the coupling direction and impact in a direction orthogonal to the coupling direction. The above-described configuration can improve the impact resistance of an electricity storage device including a plurality of cell strings. Note that the first cell string and the second cell string may be electrically connected to each other but do not have to be electrically connected to each other.

[0010] (Second Item) The electricity storage device according to the first item further includes a case that houses the first cell string and the second cell string. The case has a rectangular parallelepiped shape with a longitudinal direction oriented in the coupling direction.

[0011] When unified by the wrapping members, the first cell string and the second cell string are

easy to insert into the rectangular parallelepiped case. The rectangular parallelepiped case has a simple shape and is therefore easy to manufacture. The above-described configuration can facilitate the manufacturing of the electricity storage device and helps reduce the manufacturing cost.

[0012] (Third Item) In the electricity storage device according to the first item or the second item, the case has four faces that extend in the coupling direction and two faces that cover end portions of each of the first cell string and the second cell string. The four faces include first opposite faces that are a pair of opposite faces and second opposite faces that are opposite faces with a larger area than the first opposite faces. The wrapping members are inclined relative to the coupling direction on the first opposite faces.

[0013] Inclining the wrapping members on the faces with a smaller area as described above can reduce the likelihood of the wrapping members becoming excessively twisted.

[0014] (Fourth Item) In the electricity storage device according to any one of the first item to the third item, the wrapping members have a band shape or a sheet shape and have an adhesive face on at least one side. The adhesive face includes an adhesive agent that dissolves in an organic solvent.

[0015] This configuration helps appropriately unify the electricity storage cells of the first cell string and the electricity storage cells of the second cell string using the adhesive faces of the wrapping members. As the adhesive agent in the adhesive faces dissolves in an organic solvent, the first cell string and the second cell string are easily decomposable, which increases the recyclability of the electricity storage device.

[0016] (Fifth Item) In the electricity storage device according to any one of the first item to the fourth item, the wrapping members are each wrapped so as to straddle at least either the connection part that is located between the adjacent electricity storage cells in the first cell string or the connection part that is located between the adjacent electricity storage cells in the second cell string.

[0017] This configuration helps effectively unify the electricity storage cells of the first cell string and the electricity storage cells of the second cell string using fewer wrapping members.

[0018] Note that, as another form, a vehicle including the electricity storage device according to any one of the first item to the fifth item may be provided.

[0019] This disclosure makes it possible to improve the impact resistance of an electricity storage device including a plurality of cell strings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0021] FIG. 1 is a view for describing the configuration of an electricity storage device according to an embodiment of this disclosure;

[0022] FIG. 2 is a perspective view showing, in close-up, an inside of a case of the electricity storage device shown in FIG. 1;

[0023] FIG. 3 is a view for describing the configuration of each cell string shown in FIG. 1;

[0024] FIG. 4 is a sectional view along line IV-IV in FIG. 1;

[0025] FIG. 5 is a view showing first and second modified examples of the electricity storage device shown in FIG. 1; and

[0026] FIG. 6 is a view showing a third modified example of the electricity storage device shown in FIG. 1.

DETAILED DESCRIPTION OF EMBODIMENTS

[0027] An embodiment of this disclosure will be described in detail with reference to the drawings.

The same or equivalent parts in the drawings will be denoted by the same reference sign while description thereof will not be repeated. Of an X axis, a Y axis, and a Z axis that are orthogonal to one another in the drawings to be used below, the X axis indicates a first in-plane direction of a battery (e.g., a length direction); the Y axis indicates a second in-plane direction of the battery (e.g., a width direction); and the Z axis indicates a height direction of the battery. Hereinafter, the directions in which the arrows of the X axis, the Y axis, and the Z axis point will be denoted with a prefix of “+,” and the opposite directions will be denoted with a prefix of “-.”

[0028] FIG. 1 is a view for describing the configuration of an electricity storage device according to this embodiment. “Z-SIDE VIEW OF CASE INTERNAL STRUCTURE” in FIG. 1 is a view of contents of a case as seen from the +Z side. “Y-SIDE VIEW OF CASE INTERNAL STRUCTURE” in FIG. 1 is a view of the contents of the case as seen from the +Y side. FIG. 2 is a perspective view showing, in close-up, an end portion on the +X side of the contents of the case.

[0029] The electricity storage device according to this embodiment is a battery **100** shown in FIG. 1. The battery **100** is, for example, a secondary battery such as a lithium-ion battery, a nickel-metal hydride battery, or a sodium ion battery. Examples of lithium ion batteries include an LFP battery that adopts lithium iron phosphate as a positive electrode active material, and a ternary battery that adopts nickel-manganese-cobalt (NMC) as a positive electrode active material. The type of the secondary battery may be a liquid-type secondary battery or may be an all-solid-state secondary battery. As will be described in detail later, the battery **100** includes a plurality of electricity storage cells each functioning as a secondary battery. The battery **100** may include only electricity storage cells of the same type (e.g., only LFP batteries) or may include electricity storage cells of different types (e.g., LFP batteries and ternary batteries).

[0030] The battery **100** includes a case **300**. The case **300** has a rectangular parallelepiped external shape (rectangular parallelepiped shape) with a longitudinal direction oriented in the X direction. The case **300** has a pair of faces **F1**, **F2** opposite each other in the Z direction (first opposite faces), a pair of faces **F3**, **F4** opposite each other in the Y direction (second opposite faces), and faces **F5**, **F6** located at ends in the X direction (end faces in the X direction). The area of each of the faces **F1**, **F2** is smaller than the area of each of the faces **F3**, **F4**. The length (the dimension in the X direction) of the case **300** is longer than the width (the dimension in the Y direction) of the case **300**. The length of the case **300** may be 250 mm or longer and 5000 mm or shorter, and is, for example, about 1000 mm. The width of the case **300** may be 10 mm or longer and 1250 mm or shorter, and is, for example, about 50 mm. The ratio of the length of the case **300** to the width of the case **300** may be 4 or higher and 25 or lower. The height (the dimension in the Z direction) of the case **300** may be 10 mm or longer and 1250 mm or shorter, and is, for example, about 100 mm. However, the dimensions of the case **300** are not limited to those mentioned above.

[0031] The case **300** includes a main body **310** and a lid body **320**. The main body **310** is a housing having a tubular shape closed on one side and provided with an opening in an end face on the +X side, for example, and houses cell strings **10** and **20**. The lid body **320** is a plate-shaped member (cover member) having an external shape corresponding to the opening of the main body **310**, and closes the opening on the +X side of the main body **310**. The main body **310** and the lid body **320** may be made of the same material or may be made of different materials. As the material composing each of the main body **310** and the lid body **320**, for example, metal can be adopted. The case **300** may be an aluminum case. However, these materials can be changed as appropriate. For example, the lid body **320** may be made of an insulating material.

[0032] The cell string **10** (first cell string) includes four electricity storage cells **11** to **14** that are arrayed in the X direction and three connection parts **2A** that electrically connect the electricity storage cells to one another. The electricity storage cells **11** to **14** are connected in one row in the X direction inside the case **300**. The cell string **20** (second cell string) includes four electricity storage cells **21** to **24** that are arrayed in the X direction and three connection parts **2B** that electrically connect the electricity storage cells to one another. The electricity storage cells **21** to **24** are

connected in one row in the X direction inside the case **300**. Thus, the cell string **10** and the cell string **20** are disposed parallel to each other in the X direction. Each of the faces F1 to F4 (four faces) of the case **300** extends in a coupling direction of the cell strings **10**, **20** (X direction). Each of the faces F5, F6 of the case **300** covers end portions in the X direction of the cell strings **10**, **20**. Each electricity storage cell included in the cell strings **10** and **20** is configured to be able to store electricity.

[0033] Inside the case **300** of the battery **100**, the cell string **10** and the cell string **20** are electrically connected to each other. Specifically, as shown in FIG. 1, the end portion on the -X side (electricity storage cell **14**) of the cell string **10** and the end portion on the -X-side (electricity storage cell **24**) of the cell string **20** are electrically connected to each other inside the case **300** through a connection part **2C** having, for example, a U-shape. The connection part **2C** has basically the same structure as the connection part **2A** or **2B** except that the connection part **2C** is formed in a different shape. (While the connection part **2C** has a U-shaped cross-section, each of the connection parts **2A**, **2B** has an I-shaped cross-section.) The connection part **2C** may be an integrally molded part or may be a composite body of a plurality of parts that has been separately molded. For example, the connection part **2C** may be formed by connecting a protruding portion **144B** (FIG. 3) that protrudes from the electricity storage cell **14** and a protruding portion **144B** (FIG. 3) that protrudes from the electricity storage cell **24** to each other through an electrically conductive material (beam portion). Note that the protruding portions **144B** will be described later.

[0034] The cell strings **10** and **20** are disposed such that the positions of the corresponding electricity storage cells and the corresponding connection parts are aligned. The electricity storage cells **11**, **12**, **13**, **14** included in the cell string **10** respectively overlap the electricity storage cells **21**, **22**, **23**, **24** included in the cell string **20** in the Y direction. Thus, all the electricity storage cells included in the cell string **10** are disposed so as to each face one of the electricity storage cells included in the cell string **20** in the Y direction. The end portion on the +X side (electricity storage cell **11**) of the cell string **10** is connected to the lid body **320** through a connection terminal T1. The end portion on the +X side (electricity storage cell **21**) of the cell string **20** is connected to the lid body **320** through a connection terminal T2.

[0035] The lid body **320** has a sealing hole **321**, external terminals **322**, and a connector **323**. The sealing hole **321** may be a pressure adjustment hole for adjusting a pressure inside the case **300**. The sealing hole **321** has, for example, a seal structure composed of a metal cap (outside the case) and a seal member (inside the case). This seal structure can secure airtightness inside the case **300**, while allowing gas to be discharged to the outside of the case **300** through the sealing hole **321** when the pressure inside the case **300** exceeds a predetermined level. The external terminals **322** include an electrode tab **322A** that is joined (e.g., by laser welding) to the connection terminal T1 (FIG. 1) of the cell string **10**, and an electrode tab **322B** that is joined (e.g., by laser welding) to the connection terminal T2 (FIG. 1) of the cell string **20**. The electrode tabs **322A**, **322B** are electrically connected to the electricity storage cells **11**, **21**, respectively. Each of the electrode tabs **322A**, **322B** may have an insulating seal structure made of ceramic, for example, around the electrode. In this embodiment, the electrode tabs **322A**, **322B** function as a negative electrode tab and a positive electrode tab, respectively. However, without being limited thereto, the electrode tab **322B** and the electrode tab **322A** may be reversed in polarity and serve as a negative electrode tab and a positive electrode tab, respectively. The connector **323** includes, for example, an output terminal through which a detection signal showing a state inside the case **300** (e.g., a temperature of each electricity storage cell) detected by one or more sensors inside the case **300** is output to the outside of the case, and an input terminal through which a control signal is input from the outside of the case to one or more devices inside the case **300**. For example, a temperature sensor may be provided for each electricity storage cell inside the case **300**.

[0036] In the cell strings **10** and **20**, as shown in FIG. 1 and FIG. 2, the electricity storage cells **11** and **21** are unified as a band-shaped member **51** is wrapped around them; the electricity storage

cells **12** and **22** are unified as a band-shaped member **52** is wrapped around them; the electricity storage cells **13** and **23** are unified as a band-shaped member **53** is wrapped around them; and the electricity storage cells **14** and **24** are unified as a band-shaped member **54** is wrapped around them. Each of the members **51** to **54** is wrapped around the electricity storage cell of the cell string **10** and the electricity storage cell of the cell string **20** that are opposite each other in the Y direction, with an axis slightly inclined relative to the X direction as a rotational axis, to thereby unify these electricity storage cells. Each of the members **51** to **54** is inclined relative to the X direction (the coupling direction of the cell strings **10**, **20**) on each of a face on the $-Z$ side (the side of the face **F1**) of each electricity storage cell and a face on the $+Z$ side (the side of the face **F2**) of each electricity storage cell. The members **51** to **54** are wrapped in the same direction (wrapping direction). Each of the members **51** to **54** corresponds to one example of the “wrapping member” according to this disclosure.

[0037] The cell strings **10** and **20** are inserted into the main body **310** in a state of being wrapped with the members **51** to **54**. When unified by the members **51** to **54**, the cell strings **10** and **20** are easy to insert into the main body **310**. The members **51** to **54** may be mounted before welding of the connection parts **2A**, **2B** (e.g., welding of the protruding portions **144A** and **144B** to be described later), or may be mounted after the welding. After the cell strings **10** and **20** in the state of being wrapped with the members **51** to **54** are inserted into the main body **310**, the main body **310** and the lid body **320** are joined together. The main body **310** and the lid body **320** are welded together, for example, by laser.

[0038] Note that at least either a pressure adjustment hole or a gas discharge valve may be provided in an end face on the $-X$ side of the main body **310**. As in the end face on the $+X$ side of the main body **310**, an opening may be formed in the end face on the $-X$ side of the main body **310** as well. A lid body that has been formed separately from the tubular main body **310** may be joined to that opening (e.g., by laser welding).

[0039] In this embodiment, the cell string **10** and the cell string **20** have basically the same configuration. Hereinafter, therefore, when no distinction is made among the electricity storage cells **11** to **14** and **21** to **24**, these cells will be referred to as “electricity storage cells **1**,” and when no distinction is made between the connection parts **2A** and **2B**, these connection parts will be referred to as “connection parts **2**.”

[0040] FIG. **3** is a view for describing the configuration of each of the cell strings **10** and **20**. As shown in FIG. **3**, each cell string includes four electricity storage cells **1**. The connection parts **2** are provided between the adjacent electricity storage cells **1**, and the connection parts **2** electrically connect these electricity storage cells **1** to one another. Each cell string is formed as the electricity storage cells **1** and the connection parts **2** are alternately arrayed. In each of the cell strings **10**, **20**, the electricity storage cells **1** are connected to one another through the connection parts **2**. The rigidity of the connection part **2** is lower than the rigidity of the electricity storage cell **1**. The electricity storage cells **11** to **14** and **21** to **24** are formed by the same electricity storage cells **1**. Forming the cell strings **10** and **20** using the common electricity storage cells **1** facilitates the manufacturing of the battery **100** and can reduce the manufacturing cost.

[0041] In this embodiment, the electricity storage cell **1** is a laminate cell having one or more rolls. In the laminate cell, the one or more rolls functioning as electrode bodies are covered by a laminate exterior. In FIG. **2**, the electricity storage cells are shown with the laminate exteriors omitted. The roll has, for example, a structure in which a positive electrode sheet and a negative electrode sheet are rolled with a separator interposed therebetween. Each of the positive electrode sheet and the negative electrode sheet includes an electrode foil and an active material layer.

[0042] In the following, the structures of the electricity storage cell **1** and the connection part **2** will be described using the sectional view of FIG. **3** (an X-Y sectional view around the connection part **2**). The electricity storage cell **1** includes two rolls **110A**, **110B**, spacers **120A**, **120B**, terminal members **130A**, **130B**, and covers **150A**, **150B**.

[0043] The rolls **110A**, **110B** respectively have coated portions **111A**, **111B**, electrode tabs **112A**, **112B**, and electrode tabs **113A**, **113B**. Each of the coated portions **111A**, **111B** is a region of the electrode foil in the positive electrode sheet or the negative electrode sheet where the active material layer is provided. Each of the electrode tabs **112A**, **112B**, **113A**, **113B** is a region of the positive electrode sheet or the negative electrode sheet where the electrode foil is exposed (an uncoated portion where the active material layer is not provided). The electrode tabs **112A**, **112B** are located at end portions on the +X side of the rolls **110A**, **110B**, respectively. The electrode tabs **113A**, **113B** are located on end portions on the -X side of the rolls **110A**, **110B**, respectively.

[0044] The electrode tab **112A** and the electrode tab **112B** are disposed so as to overlap in the Y direction, and the spacer **120A** and the terminal member **130A** are provided between the electrode tab **112A** and the electrode tab **112B**. The electrode tab **113A** and the electrode tab **113B** are disposed so as to overlap in the Y direction, and the spacer **120B** and the terminal member **130B** are provided between the electrode tab **113A** and the electrode tab **113B**.

[0045] Each of the spacers **120A**, **120B** includes an insulating material (e.g., synthetic resin) and has an insulating property. Each of the spacers **120A**, **120B** has a shape of which the dimension in the Y direction becomes larger with increasing distance from the coated portions **111A**, **111B** increases. The terminal member **130A** is connected to an end face on the +X side of the spacer **120A**. The terminal member **130B** is connected to an end face on the -X side of the spacer **120B**. Each of the terminal members **130A**, **130B** includes an electrically conductive material (e.g., metal such as aluminum or copper) and has electrical conductivity. The roll **110A** and the roll **110B** are joined together (e.g., by laser welding) through the terminal members **130A** and **130B**.

[0046] Each of current collector terminals **140A**, **140B** is a component constituting a part of the connection part **2**. The current collector terminals **140A**, **140B** respectively have support portions **142A**, **142B** and the protruding portions **144A**, **144B**. One of the current collector terminals **140A** and **140B** functions as a positive-electrode current collector terminal while the other one functions as a negative-electrode current collector terminal. In one example, the positive-electrode current collector terminal is made of aluminum and the negative-electrode current collector terminal is made of copper.

[0047] Each of the current collector terminals **140A**, **140B** is formed in an L-shape, and may be formed as two plate members that have been separately molded are joined together, or may be molded in an integral state by bending process. The support portion **142A** is joined (e.g., by laser welding) to an end face on the +X side of the terminal member **130A**. The support portion **142B** is joined (e.g., by laser welding) to an end face on the -X side of the terminal member **130B**.

[0048] The cover **150A** covers an end portion on the +X side (including the electrode tabs **112A** and **112B**) of the electricity storage cell **1**. The cover **150A** is provided with a through-hole **h1** for the protruding portion **144A**. The protruding portion **144A** passes through the through-hole **h1** and protrudes to the +X side of the electricity storage cell **1**. The cover **150B** covers an end portion on the -X side (including the electrode tabs **113A** and **113B**) of the electricity storage cell **1**. The cover **150B** is provided with a through-hole **h2** for the protruding portion **144B**. The protruding portion **144B** passes through the through-hole **h2** and protrudes to the -X side of the electricity storage cell **1**.

[0049] At the connection part **2**, the protruding portion **144A** of one of the two adjacent electricity storage cells **1** is joined (e.g., by laser welding) to the protruding portion **144B** of the other electricity storage cell **1**. This welded portion may be protected by a tape or the like. On surfaces of the two rolls **110A** and **110B**, the laminate exterior **160** is provided. The laminate exterior **160** is a laminate film, for example, and is provided on a surface of the electricity storage cell **1**.

[0050] Note that the above-described configuration is merely one example of the configuration of the electricity storage cell **1**, and changes can be made thereto as appropriate. For example, the number of the rolls included in the electricity storage cell **1** is not limited to two, and may be one or three or more. As the electrode body, a stack (e.g., a stack in which a positive electrode sheet and a

negative electrode sheet are stacked with a separator interposed therebetween) may be adopted instead of a roll.

[0051] FIG. 4 is a sectional view along line IV-IV in FIG. 1. Line IV-IV lies along the wrapping direction of each of the members 51 to 54. The wrapping direction is inclined relative to the X direction in an X-Y plane. As shown in FIG. 4, the electricity storage cells 13 and 23 are unified as the member 53 is wrapped around them. While FIG. 4 shows only the member 53 provided on the pair of electricity storage cells 13 and 23 as a representative, the other pairs of electricity storage cells opposite each other in the Y direction (the pair of electricity storage cells 11, 21, the pair of electricity storage cells 12, 22, and the pair of electricity storage cells 14, 24) are similarly unified as the wrapping member (one of the members 51, 52, 54) is wrapped around them.

[0052] When the electricity storage cells of the cell string 10 and the electricity storage cells of the cell string 20 are unified as described above, the impact resistance of the cell strings 10 and 20 is likely to improve. As the wrapping members are wrapped around the electricity storage cells of the respective cell strings so as to be inclined relative to the coupling direction, the cell strings 10 and 20 are likely to have high strength against both impact in the coupling direction and impact in a direction orthogonal to the coupling direction. In addition, when the wrapping members are obliquely wrapped around the electricity storage cells of the cell string 10 and the electricity storage cells of the cell string 20, the rigidity of the cell strings 10 and 20 is likely to improve owing to the effect of the wrapping members being pliable. Thus, the above-described configuration can improve the impact resistance of an electricity storage device including a plurality of cell strings. Note that, while the electricity storage cell of the cell string 10 and the electricity storage cell of the cell string 20 next to each other in the Y direction are in contact with each other in the example shown in FIG. 4, these electricity storage cells may be disposed so as to be spaced apart.

[0053] On an inner surface of the main body 310 of the case 300, for example, an insulation layer 3 including resin, such as polyethylene terephthalate (PET), is provided. Thus, the case 300 and the parts inside the case 300 are electrically insulated from each other. However, the insulation layer 3 can be omitted in a battery in which sufficient insulation is secured. Each of the members 51 to 54 may be bonded to the inner surface of the case 300 by a bonding agent. There is a region R between a face on the +Z side of each electricity storage cell included in the cell strings 10 and 20 and the inner surface (ceiling surface) of the main body 310. In the region R, at least one of a heat management system (e.g., a heater and/or a temperature sensor), a gas discharge system (e.g., a gas flow passage and/or a pressure sensor), a flexible printed circuit board (FPC), and a wire leading to the connector 323 may be provided. A device and/or a sensor provided in the region R may be connected to the connector 323 of the lid body 320.

[0054] As shown in FIG. 1 and FIG. 2, in the battery 100, the wrapping members (members 51 to 54) are inclined relative to the coupling direction (X direction) on the first opposite faces (faces F1, F2: X-Y plane) having a smaller area than the second opposite faces (faces F3, F4: X-Z plane). Thus, the wrapping members are less likely to become excessively twisted. However, without being limited thereto, the wrapping members may be inclined on the second opposite faces instead of or in addition to the first opposite faces.

[0055] Each of the members 51 to 54 is a tape having adhesiveness, for example, on one side. Each of the members 51 to 54 is fixed to the corresponding electricity storage cells through the face having adhesiveness (adhesive face). This promotes the unification of the electricity storage cells of the cell string 10 and the electricity storage cells of the cell string 20. The adhesive face includes an adhesive agent (a substance that imparts adhesiveness to the adhesive face) that dissolves in an organic solvent. Thus, the cell strings 10 and 20 are easily decomposable, which increases the recyclability of the battery 100. In a form in which the battery 100 is a liquid-type battery, it is preferable that the members 51 to 54 be made of a material that has resistance to an electrolytic solution.

[0056] Each of the members **51** to **54** may be a tape having adhesiveness on both sides. In the case of such double-sided tapes, the cell strings **10** and **20** wrapped with the members **51** to **54** can be easily fixed to the case **300** without the need to separately provide a bonding agent. Further, a device and/or a sensor provided in the region R can be easily fixed by the adhesiveness of the double-sided tapes. In addition, the laminate exterior **160** can be easily peeled off by the double-sided tapes, which improves the recyclability of the battery **100**.

[0057] FIG. 5 is a view showing the battery **100** along with first and second modified examples of the battery **100**. In the battery **100**, an angle $\theta 1$ (more specifically, a minor angle) formed in an X-Y plane by the X direction (the coupling direction of the cell strings **10**, **20**) and the wrapping direction of each of the members **51** to **54** is smaller than 90° , and may be, for example, 25° or larger and 85° or smaller. To strengthen the unity between the electricity storage cells of the cell string **10** and the electricity storage cells of the cell string **20**, it is preferable that the angle $\theta 1$ be set to 50° or larger and 85° or smaller. A groove in which the wrapping member (one of the members **51** to **54**) enters (in other words, a groove that reduces the likelihood of displacement of the wrapping member) may be formed in a surface (at a wrapped portion) of each electricity storage cell to be wrapped with the wrapping member.

[0058] It is not essential that the wrapping directions of the respective members **51** to **54** match. The wrapping directions of the respective members **51** to **54** can be changed as appropriate. For example, the battery **100** may be modified like a battery **100A** shown in FIG. 5. The battery **100A** according to the first modified example has basically the same configuration as the battery **100**. However, the battery **100A** has a member **52A** instead of the member **52** and a member **54A** instead of the member **54**. The members **52A**, **54A** are wrapped in different directions from the members **52**, **54**, respectively. In the battery **100A**, in the X-Y plane shown in FIG. 5, a minor angle (angle $\theta 1$) between the X direction (coupling direction) and the wrapping direction of each of the members **51**, **53** is formed on the +X side, while a minor angle (angle $\theta 2$) between the X direction (coupling direction) and the wrapping direction of each of the members **52A**, **54A** is formed on the -X side. Each of the members **52A**, **54A** is wrapped so as to be inclined in the opposite direction from the members **51**, **53** with respect to a Y-Z plane (a plane orthogonal to the X direction). The magnitude of the angle $\theta 2$ may be the same as the angle $\theta 1$ or may be different from the angle $\theta 1$.

[0059] In each of the batteries **100**, **100A**, the wrapping members are band-shaped members. However, without being limited thereto, the wrapping members may be linear or sheet-shaped members. The wrapping members are not limited to tapes. For example, resin members, heat-shrinkable members, and elastic bodies (e.g., rubber bands), etc. can also be adopted as the wrapping members. The battery **100** may be modified like a battery **100B** shown in FIG. 5.

[0060] The battery **100B** according to the second modified example has basically the same configuration as the battery **100**. However, the battery **100B** includes a member **51B** instead of the members **51** and **52** and a member **52B** instead of the members **53** and **54**. The member **51B** is wrapped so as to straddle the connection part **2A** located between the adjacent electricity storage cells **11** and **12** in the cell string **10** and the connection part **2B** located between the adjacent electricity storage cells **21** and **22** in the cell string **20**. The member **52B** is wrapped so as to straddle the connection part **2A** located between the adjacent electricity storage cells **13** and **14** in the cell string **10** and the connection part **2B** located between the adjacent electricity storage cells **23** and **24** in the cell string **20**. This configuration helps effectively unify each electricity storage cell included in the cell strings **10** and **20** using fewer wrapping members.

[0061] Each of the members **51B**, **52B** may be a resin film (e.g., a film including vinyl chloride resin, polyvinylidene chloride, polyethylene, or polyolefin). Each of the members **51B**, **52B** may be a heat-shrinkable sheet.

[0062] FIG. 6 is a view showing a third modified example of the battery **100**. As shown in FIG. 6, in a battery **100C** according to the third modified example, the cell string **20** is disposed so as to be offset relative to the cell string **10** toward the +X side by an offset amount D. Thus, all the

connection parts **2A** included in the cell string **10** each overlap one of the electricity storage cells **21** to **24** of the cell string **20** in the Y direction, and all the connection parts **2B** included in the cell string **20** each overlap one of the electricity storage cells **11** to **14** of the cell string **10** in the Y direction. This configuration improves the impact resistance of the battery **100C** as each connection part is reinforced by the electricity storage cell.

[0063] The cell strings **10** and **20** are electrically connected to each other through a connection part **2D**. The connection part **2D** includes a beam portion that is formed in a plate shape in a Y-Z plane, and a first leg portion and a second leg portion that are formed in a plate shape in an X-Z plane. The first leg portion is connected to the end portion on the -X side (electricity storage cell **14**) of the cell string **10**. The second leg portion is connected to the end portion on the -X side (electricity storage cell **24**) of the cell string **20**. The beam portion is a portion that connects the first leg portion and the second leg portion to each other. The connection part **2D** is formed in a U-shape, and has basically the same structure as the connection part **2C** (FIG. 1). In the connection part **2D**, however, the length of the second leg portion in the X direction is longer than the length of the first leg portion in the X direction by the offset amount D.

[0064] The battery **100C** includes members **51C** to **54C** instead of the members **51** to **54**. Each of the members **51C** to **54C** is wrapped around the electricity storage cell of the cell string **10** and the electricity storage cell of the cell string **20** that are opposite each other in the Y direction, and is inclined in an X-Y plane relative to the X direction (the coupling direction of the cell strings **10**, **20**). Thus, the pair of electricity storage cells **11**, **21**, the pair of electricity storage cells **12**, **22**, the pair of electricity storage cells **13**, **23**, and the pair of electricity storage cells **14**, **24** are each unified. Each of the members **51C** to **54C** has an inclination angle according to a positional relationship (offset) between the electricity storage cell of the cell string **10** and the electricity storage cell of the cell string **20**. The wrapping direction of each of the members **51C** to **54C** is determined according to, for example, the offset amount D. This helps appropriately unify the cell strings **10** and **20**. As the cell strings **10** and **20** are unified in this way, the impact resistance of the battery **100C** improves.

[0065] The above-described batteries **100** and **100A** to **100C** can also independently function as an electricity storage device. However, a plurality of such batteries may be combined into a module. The batteries **100** and **100A** to **100C** can be installed, for example, in mobile bodies. Examples of mobile bodies include automobiles (battery electric vehicles, hybrid electric vehicles, etc.), vehicles other than automobiles (ships, airplanes, etc.), mobile machines (agricultural machines, construction machines, etc.), and unmanned mobile bodies (unmanned transport vehicles, robots, etc.). However, the purpose of the electricity storage device is arbitrary and may be a stationary purpose.

[0066] The number of the cell strings housed in the case **300** is not limited to two and may be three or more. Three or more cell strings may be unified by wrapping members.

[0067] The configuration of each cell string is not limited to the configuration shown in FIG. 3. Each cell string may include electricity storage cells of different dimensions, or may include electricity storage cells of different shapes. The number of the electricity storage cells housed in the case **300** is not limited to eight, either, and can be changed as appropriate. The number of the electricity storage cells included in each cell string may be smaller than four. For example, in the battery **100B** shown in FIG. 5, the cell string **20** may be composed of two electricity storage cells and one connection part. For example, the electricity storage cells **21** and **22** may be combined to form a large electricity storage cell, and the electricity storage cells **23** and **24** may be combined to form a large electricity storage cell. In such a modified example, each of the members **51B**, **52B** is wrapped so as to straddle only the connection part **2A** of the cell string **10**, without straddling the connection part **2B** of the cell string **20**. The number of the electricity storage cells included in each cell string may be five or larger and 19 or smaller or may be 20 or larger.

[0068] The various characteristics relating to the above-described electricity storage device (the

characteristics described in the embodiment and the modified examples) may be implemented in arbitrary combinations. The electricity storage device may be applied to a device other than a vehicle.

[0069] The embodiment disclosed this time should be construed as being in every respect illustrative and not restrictive. The scope of the present disclosure is indicated not by the description of the embodiment given above but by the claims, and is intended to include all changes within the meaning and scope of the claims and equivalents thereof.

Claims

1. An electricity storage device comprising a first cell string and a second cell string, wherein: the first cell string and the second cell string each include a plurality of electricity storage cells that is arrayed in a coupling direction and connection parts that electrically connect the electricity storage cells to one another; and the electricity storage device further comprises wrapping members that are inclined relative to the coupling direction at least at one part and that are wrapped around the electricity storage cells included in the first cell string and the electricity storage cells included in the second cell string.
 2. The electricity storage device according to claim 1, wherein: the electricity storage device further comprises a case that houses the first cell string and the second cell string; and the case has a rectangular parallelepiped shape with a longitudinal direction oriented in the coupling direction.
 3. The electricity storage device according to claim 2, wherein: the case has four faces that extend in the coupling direction and two faces that cover end portions of each of the first cell string and the second cell string; the four faces include first opposite faces that are a pair of opposite faces and second opposite faces that are opposite faces with a larger area than the first opposite faces; and the wrapping members are inclined relative to the coupling direction on the first opposite faces.
 4. The electricity storage device according to claim 1, wherein the wrapping members have a band shape or a sheet shape and have an adhesive face on at least one side, and the adhesive face includes an adhesive agent that dissolves in an organic solvent.
 5. The electricity storage device according to claim 1, wherein the wrapping members are each wrapped so as to straddle at least either the connection part that is located between the adjacent electricity storage cells in the first cell string or the connection part that is located between the adjacent electricity storage cells in the second cell string.
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